Is it a planet or not (false positive)?

ML Pipeline

1. **Data download**: raw light curve using lightkurve python package from MAST (Kepler data with KOI labels of host star properties)
2. **Preprocessing**: Cleaning the data, folding on period, binning the data
3. **Features**: transit depth, transit shape, odd-even depth, secondary eclipse
4. **Collinearity**: Correlation Matrix
5. **Train/Validation/Test data split**: grouping using host star
6. **Models**: Logistic regression, Decision tree, Naive Bayes, Feed Forward Network
7. **K-fold cross validation**: say, 5 fold on the training set
8. Confusion matrix, F1 score, ROC-AUC

Outline (using suggested format):

1. Main idea/question the model is being trained for
2. Pipeline Diagram (connected boxes?)
3. Data explanation
4. .ipynb notebook in Github
5. Equations with paragraph explanation
6. My own created equation metric
7. Model's learning figures
8. Model's final performance plots (ROC, Confusion matrix)
9. 4 Model comparison in a table
10. Discussion on how well it worked and result justification
11. References